

Republic of Turkey
Kastamonu University
Faculty of Engineering and Architecture
Department of Computer Engineering



DATABASE MANAGEMENT SYSTEMS
PROJECT REPORT

Project Owner	İrem AKGÜN
School Number	234410151
Advisor	Ahmet Nusret ÖZALP
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Risk Estimation from Patient History

AI-Powered Patient Intake Automation: A Google Sheets and Gemini-Based Clinical Risk Assessment Workflow

Abstract—This paper presents an automated patient intake and clinical risk assessment system built using the n8n workflow automation platform, Google Sheets, and the Google Gemini large language model. The proposed system monitors a patient registration spreadsheet in real time, triggering an AI-powered risk analysis whenever a new patient record is submitted. Patient complaints, blood sugar values, and age data are automatically forwarded to Gemini, which generates a concise risk evaluation. The result is written back to the same spreadsheet without any manual intervention. The system demonstrates how low-code automation tools can be combined with state-of-the-art language models to support clinical decision-making in resource-constrained environments.

Keywords—clinical decision support, workflow automation, large language model, Google Gemini, patient intake, n8n, Google Sheets, risk assessment

I. INTRODUCTION

The digitization of clinical workflows has become a central objective in modern healthcare informatics. Timely risk stratification of newly admitted patients can significantly affect clinical outcomes, yet many healthcare facilities rely on manual review processes that are prone to delays and human error [1]. Recent advances in large language models (LLMs) and low-code integration platforms offer an opportunity to automate routine triage tasks without requiring specialized programming expertise.

This paper describes the design and implementation of an event-driven patient intake automation system. The system leverages n8n, an open-source workflow automation tool, to orchestrate data flow between a Google Sheets patient registration form and the Google Gemini generative AI model. When a clinician or administrative staff member enters a new patient record, the system automatically extracts the relevant clinical fields, constructs a structured prompt, and invokes the Gemini API to obtain a brief risk assessment. The assessment is subsequently written back to the originating spreadsheet, making it immediately visible to clinical staff.

The remainder of this paper is organized as follows. Section II reviews related work on automated triage and LLM-based clinical support. Section III describes the system architecture. Section IV details the workflow implementation. Section V presents a discussion of the results, and Section VI concludes the paper.

II. RELATED WORK

A. Automated Triage Systems

Early automated triage systems relied on rule-based expert systems that encoded clinical guidelines as conditional logic [2]. While effective in narrow domains, these systems required extensive manual knowledge engineering and were difficult to maintain as guidelines evolved. Machine-learning approaches subsequently emerged, using structured electronic health record (EHR) data to predict patient acuity [3].

Recent work has explored the integration of natural language processing (NLP) to handle free-text chief complaints, which constitute a significant fraction of emergency department intake data [4]. Transformer-based models have demonstrated strong performance on triage classification benchmarks, achieving accuracy levels comparable to experienced nurses in controlled evaluations [5].

B. LLMs in Clinical Decision Support

The emergence of instruction-following LLMs such as GPT-4 and Google Gemini has opened new possibilities for clinical text understanding and generation. Singhal et al. [6] demonstrated that a medically fine-tuned LLM could achieve expert-level performance on medical licensing examinations. Subsequent studies have investigated zero-shot and few-shot prompting strategies for clinical note summarization and differential diagnosis generation.

Despite these advances, deployment of LLMs in live clinical workflows remains limited, primarily due to concerns regarding reliability, regulatory compliance, and integration complexity. Our work addresses the integration dimension by embedding an LLM call within a no-code automation pipeline.

III. SYSTEM ARCHITECTURE

The proposed system consists of three principal components interconnected through the n8n workflow engine: (1) a Google Sheets data source acting as a patient registration form, (2) the Google Gemini language model serving as the AI inference engine, and (3) a Google Sheets write-back mechanism that records the generated risk assessment alongside the original patient record.

TABLE I. System Component Overview

Component	Technology	Role
Trigger	Google Sheets Trigger (n8n)	Detects new patient row additions
Inference	Google Gemini (gemini-3-flash)	Generates clinical risk analysis text
Storage	Google Sheets (n8n Append/Update)	Writes AI output back to patient record

Orchestration	n8n Workflow Engine	Coordinates all node executions
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Data fields captured per patient include: full name (Hasta_Ad_Soyad), age (Yas), blood sugar level (Seker_Degeri), chief complaints (Sikayetler), and registration timestamp (Kayit_Tarihi). An additional field, Gemini_Risk_Analizi, is reserved for the AI-generated assessment and is populated automatically upon workflow execution.

IV. WORKFLOW IMPLEMENTATION

A. Trigger Node

The first node in the n8n workflow is a Google Sheets Trigger configured to poll the target spreadsheet every minute. The trigger monitors the sheet named Sayfa1 within the workbook identified by document ID 1F8dNuzeNYan60Nf8DISp96VORIk-K2zVJ5rulR_oE2M. It is set to fire on the rowAdded event, meaning that each newly appended row initiates a fresh workflow execution. OAuth 2.0 credentials are used to authenticate the trigger with the Google Sheets API.

B. AI Inference Node

The second node invokes the Google Gemini language model (gemini-3-flash-preview) via the n8n LangChain Gemini integration. A dynamic prompt is constructed by interpolating three patient fields extracted from the trigger output: Sikayetler (chief complaints), Seker_Degeri (blood sugar value), and Yas (age). The prompt instructs the model to act as a physician assistant and evaluate potential risks in one to two sentences, returning only the assessment text. Retry-on-failure is enabled with a maximum of five attempts and a four-second inter-attempt delay to handle transient API errors gracefully.

The prompt template, translated from Turkish, reads as follows: "Analyze the following patient data as a physician assistant: Patient Complaint: [complaints]; Blood Sugar Value: [value]; Age: [age]. Analysis Note: Please evaluate potential risks based on these data in 1-2 sentences and return only this evaluation."

C. Write-Back Node

The third node uses the n8n Google Sheets node in appendOrUpdate mode to persist the AI output. The node matches on the Hasta_Ad_Soyad column to locate the correct row and writes the Gemini response, extracted from the JSON path content.parts[0].text, to the Gemini_Risk_Analizi column. All other patient fields are preserved. The matching strategy ensures that concurrent executions for different patients do not overwrite each other.

V. DISCUSSION

A. Strengths

The proposed architecture offers several practical advantages. First, because n8n is a low-code platform, clinical administrators can inspect, modify, and extend the workflow without deep programming knowledge. Second, by using Google Sheets as the data layer, the system integrates natively with tools

already familiar to many healthcare staff. Third, the retry mechanism in the inference node adds resilience against the occasional latency spikes characteristic of hosted LLM APIs.

B. Limitations and Future Work

The current implementation has notable limitations. The polling interval of one minute introduces a latency that may be unacceptable in high-acuity emergency settings; a webhook-based trigger would reduce this to near-zero. The Gemini model is prompted in Turkish, which may affect output quality for non-Turkish speakers and complicates multilingual deployment. Furthermore, the system currently lacks an auditing mechanism to record model version, prompt version, and inference timestamp alongside each generated assessment.

Future work will explore: (1) integration with hospital information systems via HL7 FHIR interfaces, (2) structured output formatting using JSON schema constraints to enable downstream processing, (3) a clinician feedback loop to fine-tune prompts based on assessment accuracy, and (4) privacy-preserving deployment using on-premises LLM serving to satisfy HIPAA and GDPR requirements.

VI. CONCLUSION

This paper presented a practical patient intake automation system that combines the n8n workflow engine, Google Sheets, and the Google Gemini LLM to provide real-time clinical risk assessments without manual intervention. The system demonstrates that modern low-code automation platforms, when coupled with capable language models, can meaningfully reduce the administrative burden on clinical staff while delivering actionable decision-support information. The open and extensible design of the proposed system provides a foundation for richer clinical AI applications in resource-constrained healthcare environments.

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